

# [SHIELD] IGNISYL - ML Model Performance Report

|                   |                                        |
|-------------------|----------------------------------------|
| Report Generated: | 2025-12-26 08:47:12                    |
| Report Type:      | Machine Learning Performance Analysis  |
| Models Evaluated: | Isolation Forest, XGBoost, Autoencoder |
| Classification:   | TECHNICAL                              |

## Executive Summary

This report provides comprehensive analysis of the IGNISYL machine learning models used for insider threat detection. The system employs an ensemble approach using Isolation Forest, XGBoost Classifier, and Autoencoder neural network. Current overall accuracy is 94.2% with a detection latency of 25ms.

## 1. Overall Performance Metrics

| Metric              | Value | Target  | Status |
|---------------------|-------|---------|--------|
| Overall Accuracy    | 94.2% | > 90%   | ✓ PASS |
| False Positive Rate | 5.00% | < 10%   | ✓ PASS |
| False Negative Rate | 3.00% | < 5%    | ✓ PASS |
| Detection Latency   | 25ms  | < 100ms | ✓ PASS |
| Models Active       | 3     | 3       | ✓ PASS |
| Precision           | 92.8% | > 90%   | ✓ PASS |
| Recall              | 89.5% | > 85%   | ✓ PASS |
| F1 Score            | 91.1% | > 88%   | ✓ PASS |

## 2. Individual Model Performance

| Model            | Accuracy | Precision | Recall | F1 Score | Latency |
|------------------|----------|-----------|--------|----------|---------|
| Isolation Forest | 91.3%    | 89.7%     | 88.2%  | 88.9%    | 8ms     |



|                       |       |         |
|-----------------------|-------|---------|
| after_hours_access    | 0.065 | ■■■■■■■ |
| sensitive_file_access | 0.054 | ■■■■■■■ |
| email_attachments     | 0.031 | ■■■     |

## 6. Recommendations for Model Improvement

- Consider increasing training data for threat samples to address class imbalance.
- Evaluate adding temporal features to capture time-series patterns in user behavior.
- Test LSTM or Transformer architectures for sequence-based anomaly detection.
- Implement periodic model retraining (weekly) to adapt to evolving threat patterns.
- Add ensemble voting weights optimization using genetic algorithms.
- Consider federated learning approach for privacy-preserving model updates.
- Implement A/B testing framework for safe model deployment.
- Add explainability features (SHAP values) for security analyst review.